EXP:7 - UNCONSTRAINED OPTIMIZATION- NONLINEAR LEAST SQUARES

AIM:

The objective of this lab is to understand and implement optimization techniques

for solving nonlinear least squares problems without constraints.

PROCEDURE:

a. Problem Formulation:

- Define the nonlinear least squares objective function f(x)f(x)f(x).
- Specify the data points and the model that relates to the data.

b. Choose an Optimization Library:

- Select an appropriate optimization library (e.g., scipy optimize in Python).

c. Implementing the Optimization:

- Set up the objective function to be minimized.
- Define any constraints (if the problem extends to constrained NLS).

d. Running Optimization:

- Execute the chosen optimization algorithm.
- Monitor convergence and iterate as necessary.

e. Post-optimization Analysis:

- Evaluate and interpret the results obtained from the optimization.
- Compare with initial assumptions and data.

CODE:

```
import numpy as np

from scipy.optimize import least_squares


# Model: exponential decay (cooling process)

def model(x, t):

    return x[0] * np.exp(-x[1] * t)


# Residual function

def residual(x, t, y):

    return model(x, t) - y
```

```
# Generate synthetic temperature data
np.random.seed(123)
t = np.linspace(0, 10, 50)
y_true = [100, 0.3]  # true parameters (initial temp, cooling rate)
y = model(y_true, t) + 2 * np.random.randn(50)

# Initial guess
x0 = [50, 0.1]

# Curve fitting using least squares
result = least_squares(residual, x0, args=(t, y))

print("Optimized parameters:", result.x)
```

Output:

Optimized parameters: [99.447628 0.29780034]

Result:

The given nonlinear curve fitting problem was solved successfully using Python. The optimized parameters for the exponential decay model were obtained using least squares method.